FLEET TEAM DIGEST

737MAX-FTD-28-26001

Issue Title : Fuel Tank Access Panel Leaks

Airplane Model	ATA	Minor Model(s)
737MAX	2800-00	-8, -8200, -9

Other Model(s)	ECCN
737NG	9E991

Originated Date	Last Revised Date	Created On
10/29/2026	10/29/2026	09/28/2026

Estimated Completion Date	Next Update Date
12/31/2027	01/27/2027

Description

Operators are experiencing fuel leakage events, sometimes due to fuel tank access panel nut dome cracking, which leads to operational disruptions such as delays and AOG situations. Operators continue to experience fuel leakage events and/or nut dome cracks despite using the latest gasket with the improved wire mesh design (P/Ns 65C33095-5 and 65C33161-5) developed via 737NG-SRP-57-0227.

Additionally, 737 MAX operators are experiencing fuel leakage events, with majority of the events occurring predominantly on the machined high-impact doors, which are the 3 most inboard doors on the wings. The machined high-impact doors on the 737 MAX, along with the 737-900ER, currently utilize a Cor-Ban 27L gasket.

Background

Over the years of 737 operations, multiple operators have reported nut dome cracking and/or fuel leakage incidents common to the wing fuel tank access doors. In 2008, 737NG-SRP-57-0252 was initiated to mitigate this issue, and improved gaskets P/N 65C33095-5 (10x18 doors) and P/N 65C33161 (8x18 doors) were developed and incorporated into production. Reports of fuel leaks and nut dome cracks persisted despite incorporating the redesigned gaskets. Per ISDP, there were 93 reported fuel leakage incidents from 16 operators in 2022 and 2023 combined.

Boeing introduced gaskets impregnated with Cor-Ban 27L (P/N 65C33095-2 and 65C33161-2) which was shown to have improved corrosion prevention properties (over gaskets with Aeroshell 14) through testing between LN 3107 – LN 3549 (DEC-2009). Boeing also released gaskets impregnated with Mobilegrease 33 (P/N 65C33095-3 and 65C33161-3) to mitigate corrosion issues that resulted from the use of Aeroshell 14 grease from LN 5084 to LN 5309 (SEP-2014). Boeing later released gaskets with an improved aluminum wire mesh design impregnated with Mobilegrease 33 (P/N 65C33095-5 and 65C33161-5), from LN 5310 and on (MAR-2015).

In 2023, Boeing completed an In-Service Quality Investigation into leakages from fuel tank access panels, and the root cause was found to be indeterminate. However, due to the increased number of fuel leak reports, Boeing has reopened the root cause investigation into fuel tank access panel leaks.

Status

Boeing has concluded the root cause investigation, and has identified the following root causes associated with the 737 MAX fleet.

BOEING PROPRIETARY - The information contained herein is Proprietary to The Boeing Company and shall not be reproduced or disclosed in whole or in part or used for any reason except when such user possesses direct, written authorization from The Boeing Company. Encrypted emails required for forwarding via Email.The document generated on 11/05/2026 by Fleet Team Digest application via MyBoeingFleet. This document is for reference purposes only, visit the application for all the latest and updated revisions of this document.

Copyright ©2026 Boeing. All rights reserved.

EXPORT CONTROLLED -This technology or software is subject to the U.S. Export Administration Regulations (EAR), (15 C.F.R. Parts 730-774). No authorization from the U.S. Department of Commerce is required for export, re-export, in-country transfer, or access EXCEPT to country group E:1 or E:2 countries/persons per Supp.1 to Part 740 of the EAR. ECCN : 9E991

- Leaks without nut dome cracks (737 MAX high-impact machined doors; 531AB/631AB, 531BB/631BB, and 532AB/632AB)
  - Current installation requirements are not sufficiently robust to capture all clamp up tolerance conditions.
  - Joint relaxation (grease migration)
  - Insufficient seal compression between doors and the skin, resulting in gaps.
- Leaks with nut dome cracks
  - Due to moisture ingression, trapped moisture undergoes expansion and contraction cycles (freezing and thawing), causing repeated high stress within the dome cavity.

In a separate parallel effort, Boeing is pursuing the certification of the Av-DEC gaskets as a Retrofit solution. Boeing is coordinating with the supplier, Av-DEC, on the certification testing and qualification documentation.

In addition, Boeing is implementing new pre-molded dome filler both in production and as spares for in-service airplanes. See attachment "Pre-Molded Dome Filler" as reference.

Interim Action

Operators can use the Av-DEC gaskets approved as Supplemental Type Certificate (STC) parts. However, regional regulatory agencies may impose additional approvals or restrictions.

Operators are recommended to re-torque the fasteners to specifications to mitigate fuel leakages. Operators should re-apply original installation torque using a power torque tool with a free speed of less than 20 rpm or manual torque tool a minimum of 10 minutes after initial torque following the same torque sequence. The current Cor-Ban 27L compound verbiage callout, which applies to high-impact doors using 112W8216-1 gaskets, will be removed in future AMM revision to expand the re-torque recommendation note to all doors. In addition to the existing 60 days and 12 months re-torque intervals after installation, operators may perform additional re-torque at their own discretion on interval and frequency.

While BAC 5009 allows operators to use one or two grip lengths longer or shorter than specified, Boeing recommends not using a longer grip length as that may damage the nut domes.

Operators are recommended to use a new gasket each time a fuel tank access door is removed and installed or reinstalled. However, gasket reuse is allowed if absolutely necessary, such as at an outstation, provided the reuse conditions are met in accordance with 28-AWL-16 and the used gasket is replaced within the time-limited duration specified in 28-AWL-16.

Operators should ensure that the nut domes are free of FOD and unwanted contaminants.

Operators are recommended to add grease on the countersinks (BMS 3-33 for cast doors; BMS 3-38 for high impact doors).

Operators can drain any accumulated water in the clamp ring by unscrewing the inboard and outboard most fasteners on cast doors annually, for 20 minutes or until all moisture is drained. Re-torque is not required after reinstalling the fasteners to specification. See attachment “Cast Doors Moisture Management” as reference.

Final Action

As part of the root cause investigation, Boeing has pinpointed several solutions.

- Production Installation and AMM Updates
  - New torque sequence pattern (cast doors and high impact doors)
    - Ensures preload is distributed evenly
    - Ensures the gaps are closed
    - Minimizes loss of clamp up and seal compression
  - New instructions to re-torque at least 10 minutes after the initial torquing (cast doors and high impact doors)
    - Maintains proper clamp-up force

BOEING PROPRIETARY - The information contained herein is Proprietary to The Boeing Company and shall not be reproduced or disclosed in whole or in part or used for any reason except when such user possesses direct, written authorization from The Boeing Company. Encrypted emails required for forwarding via Email.The document generated on 11/05/2026 by Fleet Team Digest application via MyBoeingFleet. This document is for reference purposes only, visit the application for all the latest and updated revisions of this document.

Copyright ©2026 Boeing. All rights reserved.

EXPORT CONTROLLED -This technology or software is subject to the U.S. Export Administration Regulations (EAR), (15 C.F.R. Parts 730-774). No authorization from the U.S. Department of Commerce is required for export, re-export, in-country transfer, or access EXCEPT to country group E:1 or E:2 countries/persons per Supp.1 to Part 740 of the EAR. ECCN : 9E991

- Recommend preventive routine drainage – unscrew the inboard and outboard most fasteners annually (cast doors)
  - Allows any accumulated moisture to exit the cavities
- New instructions to add grease on the countersink (BMS 3-33 for cast doors; BMS 3-38 for high impact doors)
  - It reduces friction, which increases repeatable preload and improves the fuel seal around the compression joint.

- Certification of Av-DEC gaskets as additional Retrofit option.
- Implementation of new pre-molded dome filler
  - Addresses fuel leaks caused by nut dome cracks

Milestones

Note: The deliverables consist of root cause investigation and Av-DEC gasket certification. Target dates are based upon currently available status and have risk of future revision.

Root Cause Investigation
Preliminary Root Cause Established: 15-Aug-2025
Testing Complete: 5-Dec-2025
Solution Selection: 12-Dec-2025
Installation Engineering and Process Updates Release: 23-Jan-2026

Av-DEC Gasket Certification
Engineering Work Statement Complete: 24-Jun-2026
Certification/Testing Initiation: 29-Sep-2026
Certification/Testing Completion: 2Q-2027
Retrofit Solution Availability: 4Q-2027

Operator Action

Operators are encouraged to provide details of their fuel tank access panel leakage findings in Fleet IdeaXchange Item ISE-28-22-41015.

If an operator encounters a fuel leak at the wing lower access door at an outstation where no replacement door is available, a possible option to return the airplane to service would be to accomplish the attached temporary repair “Access Panel Speed Tape Temporary Repair.”

Note: The attached document is not an official published repair procedure, and is intended solely for operators' reference to illustrate the details of the temporary repair. Boeing must evaluate the repair on a case-by-case basis before permitting further flights, and can only issue a No Technical Objection (NTO).

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Service Related Problem (SRP)	737NG-SRP-57-0428		
Fleet Team Economic (FTE)	ISE-28-22-41015		
Aircraft Maintenance Manual (AMM)	Task 28-11-11-400-801		
Aircraft Maintenance Manual	Task 28-11-31-400-801		

BOEING PROPRIETARY - The information contained herein is Proprietary to The Boeing Company and shall not be reproduced or disclosed in whole or in part or used for any reason except when such user possesses direct, written authorization from The Boeing Company. Encrypted emails required for forwarding via Email.The document generated on 11/05/2026 by Fleet Team Digest application via MyBoeingFleet. This document is for reference purposes only, visit the application for all the latest and updated revisions of this document.

Copyright ©2026 Boeing. All rights reserved.

EXPORT CONTROLLED -This technology or software is subject to the U.S. Export Administration Regulations (EAR), (15 C.F.R. Parts 730-774). No authorization from the U.S. Department of Commerce is required for export, re-export, in-country transfer, or access EXCEPT to country group E:1 or E:2 countries/persons per Supp.1 to Part 740 of the EAR. ECCN : 9E991

(AMM)			
Aircraft Maintenance Manual (AMM)	Task 28-11-11-400-802		

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Gasket, Wire Mesh	112W8216-1		

Related Categories

- Fleet Team IdeaXchange : Topic of discussion on the Fleet Team IdeaXchange bulletin board
- SRPs - Economic : Service Related Problems in work by Boeing that have an economic impact

Attachments:

a) Access Panel Speed Tape Temporary Repair

b) Cast Doors Moisture Management

c) Pre-Molded Dome Filler

BOEING PROPRIETARY - The information contained herein is Proprietary to The Boeing Company and shall not be reproduced or disclosed in whole or in part or used for any reason except when such user possesses direct, written authorization from The Boeing Company. Encrypted emails required for forwarding via Email.The document generated on 11/05/2026 by Fleet Team Digest application via MyBoeingFleet. This document is for reference purposes only, visit the application for all the latest and updated revisions of this document.

Copyright ©2026 Boeing. All rights reserved.

EXPORT CONTROLLED -This technology or software is subject to the U.S. Export Administration Regulations (EAR), (15 C.F.R. Parts 730-774). No authorization from the U.S. Department of Commerce is required for export, re-export, in-country transfer, or access EXCEPT to country group E:1 or E:2 countries/persons per Supp.1 to Part 740 of the EAR. ECCN : 9E991